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Implementation of Plasma Ignition System for Flare Gas Optimization in Onshore Operations

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Abstract

In alignment with ADNOC's strategic goal of achieving net zero methane emissions by 2045, a pilot project was initiated at Sahil RDS-1 Flare to evaluate the implementation of Plasma Ignition Devices. The objective was to eliminate the continuous use of fuel gas in flare pilots, enhance ignition reliability under diverse weather conditions, and reduce greenhouse gas (GHG) emissions. The Plasma Ignition System, developed by Zhenhua Oil Company, utilizes high-voltage ionization to generate an electronic flame, offering a sustainable alternative to traditional ignition methods.

The system was tested in Continuous, Auto, and Manual modes, integrated with existing instrumentation and control systems, and monitored remotely via DCS graphics. Key achievements include successful ignition across all modes, even in simulated adverse conditions, and a reduction of 130 tons of CO₂ emissions annually for the Sahil RDS flare. The pilot also demonstrated fuel gas savings of 2.28 MMSCF per year and operational cost reductions.

This technology not only improves flare system reliability and safety but also contributes significantly to ADNOC's sustainability targets. The results support the scalability of Plasma Ignition Devices across AON assets, with projected savings of over 235 MMSCF of fuel gas and 13,461 tons of CO₂ annually.

Introduction

The oil and gas industry is under increasing pressure to reduce greenhouse gas (GHG) emissions and improve operational efficiency, particularly in upstream operations where flaring remains a necessary but environmentally impactful practice. Flare systems are critical for safe disposal of excess hydrocarbons during process upsets, maintenance, or emergencies. However, traditional flare ignition systems rely on continuous fuel gas supply to maintain pilot flames, contributing to unnecessary fuel consumption and significant GHG emissions. These systems also face reliability challenges under adverse weather conditions such as high winds, rain, and extreme temperatures.

In response to these challenges, ADNOC Onshore initiated a pilot project at the Sahil RDS-1 Flare to evaluate the feasibility and performance of Plasma Ignition Devices. This initiative aligns with ADNOC's broader sustainability strategy, which includes achieving net zero methane emissions by 2045 and zero routine flaring by 2030. The project also supports ADNOC's commitment to operational excellence and innovation through the deployment of advanced technologies that reduce environmental impact while enhancing safety and reliability.

Traditional ignition systems in ADNOC flares are designed per AGES-SP-06-006 and AGES-PH-08-002 specifications, incorporating dual ignition mechanisms—High Energy Spark Ignition (HESI) and Flame Front Generator (FFG). While these systems are robust, they require continuous fuel gas supply and are sensitive to air-fuel ratios, making them less effective during abnormal weather conditions. Moreover, the need for multiple pilots (minimum four for flares ≥ 36 " diameter) and backup systems increases complexity and operational costs.

The Plasma Ignition System offers a transformative alternative. Instead of relying on fuel gas, it uses high-voltage electrical energy to ionize gas molecules and generate a high-temperature electronic flame. This eliminates the need for continuous pilot operation and significantly reduces fuel gas consumption and associated emissions. The system is designed to operate in three modes—Continuous, Auto, and Manual—each tailored to different operational scenarios. It integrates seamlessly with existing instrumentation and control systems, allowing remote monitoring and control via DCS graphics.

The pilot project at Sahil RDS-1 was executed in collaboration with Zhenhua Oil Company, which supplied and installed the Plasma Ignition Devices. The flare was selected based on ease of installation, minimal operational disruption, and availability of existing infrastructure. The project was approved under ADNOC's Tech Road Initiatives and involved stakeholders from Sahil Operations, Technical Centre, CHSE, and Zhenhua Oil. A comprehensive evaluation was conducted, including Factory Acceptance Testing (FAT), installation during a scheduled shutdown, and commissioning in February 2025.

This manuscript presents the background, methodology, results, and impact of the Plasma Ignition System pilot project. It highlights how technology addresses the limitations of conventional flare ignition systems and contributes to ADNOC's sustainability and operational goals. The findings demonstrate the system's reliability across all weather conditions, its effectiveness in reducing GHG emissions, and its potential for scalability across ADNOC Onshore assets.

Objective / Scope

The primary objective of this pilot project was to evaluate the feasibility, performance, and operational benefits of implementing Plasma Ignition Devices on the Sahil RDS-1 Flare. This initiative was driven by ADNOC's strategic commitment to achieving net zero methane emissions by 2045 and eliminating routine flaring by 2030. The project aimed to replace traditional fuel gas-based flare ignition systems with a more sustainable, electrically driven plasma ignition technology.

The scope of the project encompassed the full lifecycle of deployment—from technical feasibility assessment and stakeholder alignment to installation, commissioning, and performance monitoring. The Plasma Ignition System was tested in three operational modes: Continuous, Auto, and Manual. Each mode was evaluated under varying environmental conditions to ensure reliability and adaptability across ADNOC Onshore assets.

The implementation boundaries were defined to include integration with existing instrumentation and control systems, remote operability via DCS graphics, and minimal disruption to ongoing operations. The pilot was confined to the Sahil RDS-1 Flare, selected for its accessibility and infrastructure readiness. However, the outcomes were analyzed with scalability in mind, considering potential deployment across all AON flares.

This section outlines the specific goals and constraints of the project, setting the foundation for the detailed methodology and results discussed in subsequent sections.

Methodology

Overview of the Innovation

The Plasma Ignition System deployed at Sahil RDS-1 Flare represents a novel approach to flare ignition, replacing conventional fuel gas-based pilot systems with an electrically driven plasma flame. This system utilizes high-voltage ionization to generate a reliable ignition source, eliminating the need for continuous fuel gas supply and significantly reducing greenhouse gas (GHG) emissions. The technology was selected for its potential to enhance operational efficiency, improve ignition reliability under diverse weather conditions, and support ADNOC's sustainability goals.

Technical Architecture and Components

The Plasma Ignition System comprises four primary components:

- **Ignition Control Panel:** Manages ignition logic and power distribution.
- **Flow Detection Unit:** Monitors flare header flow and pressure to trigger ignition.
- **Ignition Unit:** Generates plasma flame using high-voltage discharge.
- **Flame Detection Unit:** UV/IR sensors confirm flame presence and ensure safety.

The system supports three operational modes:

- **Continuous Mode:** Two of three igniters operate alternately every hour to maintain readiness.
- **Auto Mode:** Igniters activate automatically upon detecting vent gas flow and pressure thresholds.
- **Manual Mode:** Igniters are triggered manually either locally or remotely from the control room.

Integration with ADNOC's ICSS (Integrated Control and Safety System) enabled remote monitoring and control via DCS graphics. Thresholds for flow and pressure were configured to trigger ignition in Auto Mode, and counters were added to track ignition events per igniter.

Implementation Phases

The project was executed in four structured phases:

Phase I – Feasibility and Planning. Stakeholder workshops were conducted with Sahil Operations, Technical Centre, CHSE, and Zhenhua Oil Company. Technical and HSE concerns were addressed, and the proposal was submitted under ADNOC's Tech Road Initiatives. Approval was granted by the Tech Road Committee.

Phase II – Procurement and Fabrication. Following a formal MOU between ADNOC Onshore and Zhenhua Oil Company, purchase orders were issued for the supply and installation of Plasma Ignition Devices. Fabrication was completed, and Factory Acceptance Testing (FAT) was witnessed by Sahil Plant Operations.

Phase III – Installation and Integration. Devices were shipped to UAE and installed during the December/January Sahil shutdown. Foundations were laid for flame detectors and control panels. Electrical cabling and power supply (230 KV, 50 Hz AC) were connected. The system was integrated with existing instrumentation and configured on DCS.

Phase IV – Commissioning and Testing. Commissioning took place in February 2025. Live demonstrations were conducted for teams from AON fields, CHSE, and Technical Centre. All three modes were tested under simulated weather conditions, including rain via water sprinklers.

Stakeholder Engagement

The pilot project was a collaborative effort involving:

- **Sahil Operations & Maintenance:** Led site preparation, installation, and monitoring.
- **Technical Centre:** Provided subject matter expertise and validation.
- **CHSE:** Ensured compliance with safety and environmental standards.
- **Zhenhua Oil Company:** Supplied technology, conducted FAT, and led installation.
- **AON Field Representatives:** Participated in demonstrations and feedback sessions.

This cross-functional collaboration ensured smooth execution and comprehensive evaluation of the technology.

Timeline and Milestones

Milestone	Date
Feasibility Workshops	Q4 2024
Tech Road Approval	November 2024
MOU and Procurement	December 2024
FAT and Shipment	January 2025
Installation and Integration	January 2025
Commissioning and Testing	February 2025
Performance Monitoring	February–May 2025

Safety and Environmental Considerations

Safety was a core focus throughout the project. The Plasma Ignition System was evaluated for:

- **Operational Safety:** UV/IR flame detectors ensure immediate response to flame loss.
- **Electrical Safety:** All components were tested for insulation and grounding compliance.
- **Environmental Impact:** The system eliminates fuel gas usage, reducing GHG emissions significantly.

The pilot demonstrated a reduction of 130 tons of CO₂ annually for the Sahil RDS flare. If scaled across all AON flares, the projected reduction is 13,461 tons of CO₂ per year.

Benchmarking

Plasma ignition systems save fuel and require less maintenance compared to conventional flare ignition systems, which use **2.28 mmscf/Y** of gas. Conventional systems are difficult to maintain due to their complex piping, instrumentation, and electrical components. Switching to plasma ignition overcomes these issues.

Scalability Challenges

System installation and scaling are straightforward, with a unit cost of **USD 230K**, significantly less than conventional systems. Installation and configuration can be completed during shutdowns without hot works. The table below shows fuel gas savings achieved by using this technology across all ADNOC fields.

Fields	Gas volume potential saving (mmscf per yr)
A	10.22
B	27.63
C	7.07
D	63.54
E	22.02
F	34.59
G	16.57
H	32.12
I	15.33
J	6.41

Future Research

ADNOC has implemented various technical solutions to reduce greenhouse gas (GHG) emissions and is working towards a net zero roadmap. The plasma ignition system is an example of a technology being used to help lower GHG emissions.

Results & Discussion

Quantitative Performance Outcomes

The pilot implementation of the Plasma Ignition System at Sahil RDS-1 Flare yielded substantial operational and environmental benefits. The system was evaluated over a multi-month period beginning in February 2025, with performance monitored across three ignition modes: Continuous, Auto, and Manual. The system demonstrated consistent ignition reliability, eliminating the need for fuel gas supply to flare pilots.

Key performance indicators (KPIs) recorded during the pilot include:

- Fuel Gas Savings: 2.28 MMSCF/year
- Cost Savings: USD 5,358/year
- GHG Emissions Reduction: 130 tons CO₂/year

These results were derived from real-time monitoring of ignition events, flare header flow rates, and combustion status. The system's ability to ignite on demand without fuel gas dependency directly contributed to ADNOC's sustainability goals.

If scaled across all AON flares, the projected annual savings are:

Metric	Sahil RDS-1 Flare	All AON Flares
Fuel Gas Savings (MMSCF)	2.28	235.51
Cost Savings (USD)	5,358	553,441
GHG Reduction (T CO ₂)	130	13,461

These figures underscore the system's scalability and its potential to deliver enterprise-wide impact.

Power Consumption Analysis

The Plasma Ignition System was powered using existing 230 KV, 50 Hz AC infrastructure. Power consumption varied slightly across operational modes due to differences in igniter activation logic and component usage.

Mode	Total Power Consumption (Watt)	Remarks
Manual	4,830	Igniters active for 60s per event
Automatic	4,830	Triggered by flow/pressure signals
Continuous	4,230	Two igniters active continuously

The system's energy efficiency was validated through DCS-integrated counters and instrumentation logs. The flame detectors and PLC control cabinet operated continuously, while air conditioners were configured to activate based on internal cabinet temperature thresholds.

Operational Reliability and Environmental Resilience

The Plasma Ignition Devices were tested under diverse environmental conditions, including high winds, rain, and temperature extremes. Rain simulation was conducted using water sprinklers to evaluate system resilience. Across all modes, the system maintained reliable ignition, with automatic re-ignition triggered upon flame loss detection.

Ignition counters recorded the following activity by May 2025:

- Pilot 1: #1,700 ignitions
- Pilot 2: #1,400 ignitions
- Pilot 3: #1,400 ignitions

The system's design life is rated at 5 years or 500,000 ignitions, with no maintenance issues reported during the pilot. This reliability supports long-term deployment with minimal operational disruption.

Comparative Evaluation: Before vs. After Implementation

A comparative analysis was conducted to assess the Plasma Ignition System against traditional flare ignition methods used in ADNOC operations.

Parameter	Traditional System	Plasma Ignition System
Fuel Gas Requirement	Continuous	None
Weather Sensitivity	High	Low
Emission Profile	High GHG	Reduced GHG
Ignition Reliability	Variable	High
Maintenance Frequency	Moderate	None (pilot phase)
Remote Operability	Limited	Full DCS Integration
Safety Redundancy	Manual FFG backup	UV/IR flame detection

The Plasma Ignition System addressed key limitations of the conventional setup, particularly in terms of fuel dependency, weather resilience, and automation. The integration with existing ICSS infrastructure enabled seamless remote monitoring and control, enhancing operational visibility and responsiveness.

Visuals and Instrumentation

The pilot included the deployment of several visual and instrumentation tools:

- Figure 1: General Principle of Plasma Ignition System
- Figure 2: Processing Flow Chart
- Figure 3: Component Layout
- **Figure 4: Igniter Counter Display**
- **Figure 5: DCS Graphic Interface**
- **Figure 6: eLog Snapshot of Ignition Events**

These visuals were used during commissioning demonstrations and stakeholder workshops to validate system functionality and facilitate feedback collection.

Lessons Learned

The pilot project provided several operational and strategic insights:

- **Flare Selection:** Choosing a flare with minimal modification requirements and accessible infrastructure was critical to project success. Sahil RDS-1 was selected for its readiness and integration potential.
- **Pre-shutdown Preparation:** Civil works, including foundation installation and cable routing, were completed ahead of the shutdown window. This proactive approach reduced implementation time and minimized disruption.
- **Stakeholder Collaboration:** Early engagement with Operations, CHSE, Technical Centre, and Zhenhua Oil ensured alignment and expedited approvals. The collaborative model enabled rapid issue resolution and knowledge sharing.
- **Instrumentation Integration:** Leveraging existing flow meters and pressure transmitters reduced cost and complexity. DCS integration allowed real-time monitoring and control, enhancing safety and efficiency.

Limitations and Mitigation Strategies

While the pilot was largely successful, certain limitations were identified:

- **Low Flow/H₂S Vent Gas Sensitivity:** In Auto Mode, the system relies on flow and pressure thresholds to trigger ignition. Concerns were raised about its performance under low-volume, high-H₂S venting conditions. To mitigate this, the system was operated in Continuous Mode during the pilot, which does not depend on flow signals.
- **Spare Parts Availability:** Feedback from field teams highlighted the need for local availability of spare parts. Recommendations were made to establish vendor support and inventory stocking within the UAE.
- **Training and Familiarization:** Although the system was intuitive, additional training modules were suggested to enhance operator confidence and ensure consistent usage across sites.

Stakeholder Feedback

Feedback was collected from Subject Matter Experts (SMEs) and field representatives during live demonstrations:

- Alagiri Chinnappan (TC – Integrity & Operational Excellence): Highlighted the system's potential and recommended further evaluation of lifespan, reliability, and power usage.
- Hamoud Abdulbari Al Kendi (NEB Operations): Emphasized the system's seamless deployment and its alignment with ADNOC's emission reduction goals. Suggested continuous mode operation for optimal performance.

This feedback was instrumental in refining deployment strategies and informing future scalability plans.

Conclusion

The successful implementation of the Plasma Ignition System at Sahil RDS-1 Flare marks a significant milestone in ADNOC's journey toward operational excellence and environmental stewardship. This pilot project demonstrated the feasibility and effectiveness of replacing traditional fuel gas-based flare ignition systems with a high-voltage plasma-based solution, aligning directly with ADNOC's strategic goals of achieving net zero methane emissions by 2045 and eliminating routine flaring by 2030.

The Plasma Ignition Devices operated reliably across all three modes—Continuous, Auto, and Manual—under diverse environmental conditions, including high winds, rain, and temperature extremes. The system's integration with existing ICSS infrastructure enabled seamless remote monitoring and control, enhancing safety and operational responsiveness. Quantitative results from the pilot included fuel gas savings of 2.28 MMSCF/year, cost reductions of USD 5,358/year, and a GHG emissions reduction of 130 tons of CO₂ annually. These outcomes validate the system's potential to deliver measurable sustainability and cost-efficiency benefits.

Beyond the immediate operational gains, the pilot project provided valuable insights into implementation strategy, stakeholder engagement, and system scalability. Lessons learned regarding flare selection, pre-shutdown preparation, and instrumentation integration will inform future deployments. The collaborative approach involving Sahil Operations, Technical Centre, CHSE, and Zhenhua Oil Company ensured alignment across technical and HSE domains, facilitating smooth execution and knowledge transfer.

The relevance of this technology extends beyond ADNOC Onshore. Its scalability across AON assets, combined with its low maintenance profile and robust performance, positions it as a viable industry-wide solution for flare optimization. If deployed across all AON flares, the projected annual savings exceed 235 MMSCF of fuel gas, USD 553,441 in operational expenditure, and 13,461 tons of CO₂ emissions—underscoring its transformative potential.

In conclusion, the Plasma Ignition System offers a compelling value proposition for ADNOC and the broader oil and gas industry. It enhances flare system reliability, reduces environmental impact, and contributes to long-term sustainability and OPEX reduction. The pilot's success paves the way for broader adoption and integration of plasma-based ignition technologies across ADNOC's operational landscape.

Recommendations / Future Work

Based on the successful pilot implementation of the Plasma Ignition System at Sahil RDS-1 Flare, several recommendations are proposed to guide future scaling, optimization, and integration efforts across ADNOC Onshore assets.

Scalability Potential

The pilot results demonstrate that the Plasma Ignition System is technically and operationally viable for deployment across multiple flare stacks. To maximize impact, it is recommended that ADNOC

prioritize flares with high fuel gas consumption and frequent ignition cycles. A phased rollout strategy should be adopted, beginning with assets that have existing infrastructure compatible with plasma ignition components, such as integrated control systems and accessible flare headers.

System Optimization

While the system performed reliably across all modes, further optimization is suggested to enhance its adaptability to low-flow, high-H₂S venting conditions. This includes refining the Auto Mode logic to improve sensitivity thresholds and incorporating predictive analytics to anticipate ignition needs based on process conditions. Additionally, power consumption can be further reduced by evaluating alternative cooling mechanisms for the control cabinet and optimizing igniter duty cycles.

Long-Term Monitoring and Integration

To ensure sustained performance and reliability, a long-term monitoring framework should be established. This includes periodic inspection of igniter components, calibration of flame detectors, and analysis of ignition event logs. Integration with ADNOC's enterprise-wide digital platforms, such as predictive maintenance dashboards and emissions tracking systems, will enhance visibility and support data-driven decision-making.

Training and Support

As the system is scaled, comprehensive training programs should be developed for operations and maintenance personnel. These should cover system functionality, troubleshooting procedures, and safety protocols. Vendor partnerships should be strengthened to ensure timely availability of spare parts and technical support within the UAE.

By implementing these recommendations, ADNOC can accelerate the transition to low-emission flare operations, reduce OPEX, and reinforce its leadership in sustainable energy practices.

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